Unit 3

Lesson Two

Vectors in the space

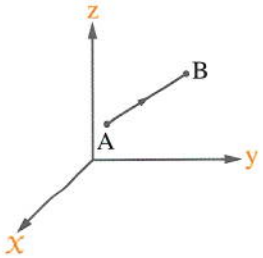
• The directed line segment :

The directed line segment is defined by three elements :

1. The starting point. **2.** The end point.

3. The direction from the starting point to the end point.

i.e. The directed line segment is a line segment which has starting point , end point and direction.

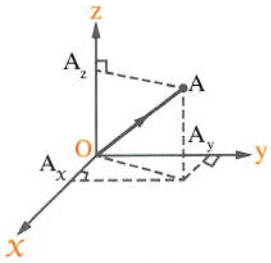


For example :

A (x_1 , y_1 , z_1) is the starting point and B (x_2 , y_2 , z_2) is the end point and its direction is the direction of $\overrightarrow{AB}$ and its norm $\| \overrightarrow{AB} \|$ is the length of $\overline{AB}$ and $\overrightarrow{AB}$ denoted both the norm and the direction.

• Position vector in the space :

The position vector of point A (A_x , A_y , A_z) with respect to the origin O $(0 , 0 , 0)$ is defined as the directed line segment whose starting point is O and end point is A and its direction $\overrightarrow{OA}$



- The position vector of point A is denoted by $\overrightarrow{OA} = (A_x , A_y , A_z)$ and since all the position vectors is relative to origin "O" so it can be written as $\vec{A}$ instead of $\overrightarrow{OA}$ to express the position vector of point A with respect to origin.
- A_x is called the component of $\vec{A}$ in direction of x-axis.
- A_y is called the component of $\vec{A}$ in direction of y-axis.
- A_z is called the component of $\vec{A}$ in direction of z-axis.

• The norm of a vector :

It is the length of the directed line segment which represents the vector.

If $\vec{A} = (A_x , A_y , A_z)$

, $\because \vec{A}$ is the directed line segment from the origin "O" to the point "A" and from the distance rule between two points.

$\therefore$ The length of $\vec{A}$, denoted as $\| \vec{A} \| = \sqrt{(A_x - 0)^2 + (A_y - 0)^2 + (A_z - 0)^2}$

i.e. $\|\overrightarrow{A}\| = \sqrt{(A_x)^2 + (A_y)^2 + (A_z)^2}$

For example :

If $\overrightarrow{A} = (-2, 1, 2)$, then $\|\overrightarrow{A}\| = \sqrt{(-2)^2 + (1)^2 + (2)^2} = 3$ length units.

- **Adding vectors in the space :**

If $\overrightarrow{A} = (A_x, A_y, A_z)$, $\overrightarrow{B} = (B_x, B_y, B_z)$, then

$\overrightarrow{C} = \overrightarrow{A} + \overrightarrow{B} = (A_x + B_x, A_y + B_y, A_z + B_z) = (C_x, C_y, C_z)$

For example :

If $\overrightarrow{A} = (-2, 3, 5)$, $\overrightarrow{B} = (4, -3, 2)$

, then $\overrightarrow{A} + \overrightarrow{B} = (-2, 3, 5) + (4, -3, 2) = (2, 0, 7)$

- **Properties of adding vectors in the space :**

For any two vectors $\overrightarrow{A}$ and $\overrightarrow{B} \in \mathbb{R}^3$, then :

1. **Closure property :** $\overrightarrow{A} + \overrightarrow{B} \in \mathbb{R}^3$
2. **Commutative property :** $\overrightarrow{A} + \overrightarrow{B} = \overrightarrow{B} + \overrightarrow{A}$
3. **Associative property :** $(\overrightarrow{A} + \overrightarrow{B}) + \overrightarrow{C} = \overrightarrow{A} + (\overrightarrow{B} + \overrightarrow{C}) = \overrightarrow{A} + \overrightarrow{B} + \overrightarrow{C}$ where $\overrightarrow{C} \in \mathbb{R}^3$
4. **Identity element over addition (zero vector) :** $\overrightarrow{O} = (0, 0, 0)$ is the neutral element of addition in $\mathbb{R}^3$

   i.e. $\overrightarrow{A} + \overrightarrow{O} = \overrightarrow{O} + \overrightarrow{A} = \overrightarrow{A}$
5. **The additive inverse :** For every vector $\overrightarrow{A} = (A_x, A_y, A_z) \in \mathbb{R}^3$ there is $(-\overrightarrow{A}) = (-A_x, -A_y, -A_z) \in \mathbb{R}^3$ such that : $\overrightarrow{A} + (-\overrightarrow{A}) = (-\overrightarrow{A}) + \overrightarrow{A} = \overrightarrow{O}$

   and the vector $(-\overrightarrow{A})$ is called the additive inverse of $\overrightarrow{A}$
6. **Elimination property :** For any three vectors $\overrightarrow{A}$, $\overrightarrow{B}$, $\overrightarrow{C}$

   If $\overrightarrow{A} + \overrightarrow{B} = \overrightarrow{A} + \overrightarrow{C}$, then $\overrightarrow{B} = \overrightarrow{C}$

Remark

If $\overrightarrow{A}$, $\overrightarrow{B}$ are two vectors in the space , then $\|\overrightarrow{A} + \overrightarrow{B}\| \leq \|\overrightarrow{A}\| + \|\overrightarrow{B}\|$

- **Multiplying a vector by a scalar (a real number) :**

If $\overrightarrow{A} = (A_x, A_y, A_z) \in \mathbb{R}^3$ and $k \in \mathbb{R}$, then :

$k\overrightarrow{A} = k(A_x, A_y, A_z) = (kA_x, kA_y, kA_z) \in \mathbb{R}^3$

For example :

If $\overrightarrow{A} = (4 , -2 , 5)$, then : • $3\overrightarrow{A} = (12 , -6 , 15)$ • $\frac{1}{2}\overrightarrow{A} = \left(2 , -1 , \frac{5}{2}\right)$

• Properties of multiplying vectors by a real number :

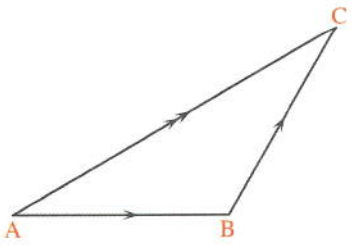
If $\overrightarrow{A} , \overrightarrow{B} \in \mathbb{R}^3$ and $k , \ell \in \mathbb{R}$, then

1. Distributive property : • $k(\overrightarrow{A} + \overrightarrow{B}) = k\overrightarrow{A} + k\overrightarrow{B}$ • $(k + \ell)\overrightarrow{A} = k\overrightarrow{A} + \ell\overrightarrow{A}$

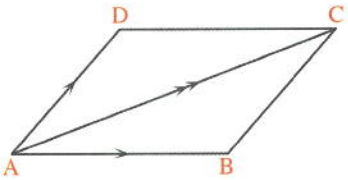
2. Associative property : • $k(\ell\overrightarrow{A}) = (k\ell)\overrightarrow{A} = k\ell\overrightarrow{A}$

3. Elimination property : If $k \neq 0$ and $k\overrightarrow{A} = k\overrightarrow{B}$, then $\overrightarrow{A} = \overrightarrow{B}$

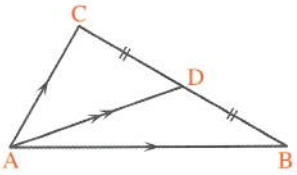
• Some geometric relations of vectors :



$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$



$\overrightarrow{AB} + \overrightarrow{AD} = \overrightarrow{AC}$



$\overrightarrow{AB} + \overrightarrow{AC} = 2\overrightarrow{AD}$

Example 1

If $\overrightarrow{A} = (3 , -1 , 2)$, $\overrightarrow{B} = (2 , 5 , -3)$, $\overrightarrow{C} = (-4 , 2 , 6)$, find each of the following vectors :

1. $3\overrightarrow{A} - 2\overrightarrow{B}$ **2.** $\frac{1}{2}(4\overrightarrow{A} - 2\overrightarrow{B} + 3\overrightarrow{C})$

3. $3\overrightarrow{O} - 2(\overrightarrow{C} + 3\overrightarrow{A})$ «where $\overrightarrow{O}$ is the zero vector»

Solution

1. $3\overrightarrow{A} - 2\overrightarrow{B} = 3(3 , -1 , 2) - 2(2 , 5 , -3) = (9 , -3 , 6) - (4 , 10 , -6) = (5 , -13 , 12)$

2. $\frac{1}{2}(4\overrightarrow{A} - 2\overrightarrow{B} + 3\overrightarrow{C}) = \frac{1}{2}[4(3 , -1 , 2) - 2(2 , 5 , -3) + 3(-4 , 2 , 6)]$

$= \frac{1}{2}[(12 , -4 , 8) + (-4 , -10 , 6) + (-12 , 6 , 18)]$

$= \frac{1}{2}(-4 , -8 , 32) = (-2 , -4 , 16)$

3. $3\overrightarrow{O} - 2(\overrightarrow{C} + 3\overrightarrow{A}) = 3(0 , 0 , 0) - 2[(-4 , 2 , 6) + 3(3 , -1 , 2)]$

$= -2[(-4 , 2 , 6) + (9 , -3 , 6)] = -2(5 , -1 , 12) = (-10 , 2 , -24)$

Example 2

If $\overrightarrow{A} = (5 , -1 , 2)$, $\overrightarrow{B} = (7 , -6 , 4)$
, find : $\| 2\overrightarrow{A} - \overrightarrow{B} \|$

Solution

$2\overrightarrow{A} - \overrightarrow{B} = 2(5 , -1 , 2) - (7 , -6 , 4)$

$= (10 , -2 , 4) - (7 , -6 , 4) = (3 , 4 , 0)$

$\therefore \| 2\overrightarrow{A} - \overrightarrow{B} \| = \sqrt{(3)^2 + (4)^2 + (0)^2} = 5$ length unit.

Example 3

If $\overrightarrow{A} = (-1 , 0 , -6)$, $\overrightarrow{B} = (3 , 3 , k + 1)$ and $\| \overrightarrow{A} + \overrightarrow{B} \| = 7$ length unit
, find the value of k

Solution

$\overrightarrow{A} + \overrightarrow{B} = (-1 , 0 , -6) + (3 , 3 , k + 1) = (2 , 3 , k - 5)$

, $\because \| \overrightarrow{A} + \overrightarrow{B} \| = 7$ $\therefore (2)^2 + (3)^2 + (k - 5)^2 = 7^2$

$\therefore 4 + 9 + k^2 - 10k + 25 = 49$ $\therefore k^2 - 10k - 11 = 0$

$\therefore (k - 11)(k + 1) = 0$ $\therefore k = 11$ or $k = -1$

Remark

If $\overrightarrow{A}$, $\overrightarrow{B}$ are two vectors in the space , m is a real number $\neq 0$ and :
$\| \overrightarrow{B} \| = \| m\overrightarrow{A} \|$, then $\| \overrightarrow{B} \| = | m | . \| \overrightarrow{A} \|$

For example :

If • $\| \overrightarrow{B} \| = \| -3\overrightarrow{A} \|$, then $\| \overrightarrow{B} \| = 3 \| \overrightarrow{A} \|$

• $\| \overrightarrow{B} \| = \| 5\overrightarrow{A} \|$, then $\| \overrightarrow{B} \| = 5 \| \overrightarrow{A} \|$

- **Equality of two vectors in the space :**

If $\overrightarrow{A} = (A_x , A_y , A_z)$, $\overrightarrow{B} = (B_x , B_y , B_z)$, then

$\overrightarrow{A} = \overrightarrow{B}$ if and only if : $A_x = B_x$, $A_y = B_y$, $A_z = B_z$

For example :

If $\vec{A} = (2 , m , n + 5)$, $\vec{B} = (\ell , -3 , 3n + 1)$ and $\vec{A} = \vec{B}$, then :

$\ell = 2$, $m = -3$

, $n + 5 = 3n + 1$ $\quad\quad$ $\therefore n = 2$

Example 4

If $\vec{A} = (2 , -1 , -3)$, $\vec{B} = (4 , 0 , 2)$, $\vec{C} = (-8 , -2 , -12)$, express $\vec{C}$ in terms of $\vec{A}$ and $\vec{B}$

Solution

Let $\vec{C} = k\vec{A} + \ell\vec{B}$ where k and $\ell \in \mathbb{R}^*$

$\therefore \vec{C} = k(2 , -1 , -3) + \ell(4 , 0 , 2) = (2k , -k , -3k) + (4\ell , 0 , 2\ell)$

$\therefore (-8 , -2 , -12) = (2k + 4\ell , -k , -3k + 2\ell)$

$\therefore -2 = -k$ $\quad\quad$ $\therefore k = 2$

, $\because 2k + 4\ell = -8$ $\quad\quad$ $\therefore 2(2) + 4\ell = -8$

$\therefore \ell = -3$ $\quad\quad$ $\therefore \vec{C} = 2\vec{A} - 3\vec{B}$

- **The unit vector :**

The unit vector is the vector whose norm equals the unit length.

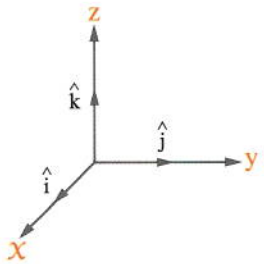
For example :

$\vec{A} = \left(\frac{3}{5} , \frac{-2}{5} , \frac{2\sqrt{3}}{5}\right)$ is a unit vector because $\|\vec{A}\| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{-2}{5}\right)^2 + \left(\frac{2\sqrt{3}}{5}\right)^2} = 1$

- **Fundamental unit vectors ($\hat{i}$, $\hat{j}$, $\hat{k}$) :**

They are directed line segments whose starting point is the origin point and their norms are the unit length and their directions are the positive directions of x , y and z axes respectively.

So we have three fundamental unit vectors :



1. The fundamental unit vector $\hat{i}$:

It is a position vector of the point $(1 , 0 , 0)$ and its norm is one unit and its direction is the positive direction of x-axis.

2. The fundamental unit vector $\hat{j}$:

It is a position vector of the point $(0\ ,1\ ,0)$ and its norm is one unit and its direction is the positive direction of y-axis.

3. The fundamental unit vector $\hat{k}$:

It is a position vector of the point $(0\ ,0\ ,1)$ and its norm is one unit and its direction is the positive direction of z-axis.

*** Notice that :** $\|\hat{i}\| = \|\hat{j}\| = \|\hat{k}\| = 1$

- **Expressing a vector in the space in terms of the fundamental unit vectors :**

Let $\vec{A}$ be a vector in 3-dimensional space $(\vec{A} \in \mathbb{R}^3)$ where

$\vec{A} = (A_x\ , A_y\ , A_z)$

$\therefore \vec{A} = (A_x\ , 0\ , 0) + (0\ , A_y\ , 0) + (0\ , 0\ , A_z)$ (from definition of addition)

$= A_x(1\ , 0\ , 0) + A_y(0\ , 1\ , 0) + A_z(0\ , 0\ , 1)$

(from definition of multiplying by real number)

$$\therefore \vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

and the previous form is used to express any vector in the space in terms of the fundamental unit vectors.

For example : $\vec{B} = (-5\ , 7\ , \sqrt{3}) = -5\hat{i} + 7\hat{j} + \sqrt{3}\hat{k}$

Example 5

If $\vec{A} = (-8\ , 5\ , 9)$, $\vec{B} = (1\ , -2\ , 3)$, express $\vec{A}$ and $\vec{B}$ in terms of the fundamental unit vectors , then find : $\vec{C} = \vec{A} - 3\vec{B}$ and calculate its norm.

Solution

$\vec{A} = -8\hat{i} + 5\hat{j} + 9\hat{k}$, $\vec{B} = \hat{i} - 2\hat{j} + 3\hat{k}$

$\vec{C} = \vec{A} - 3\vec{B} = (-8\hat{i} + 5\hat{j} + 9\hat{k}) - 3(\hat{i} - 2\hat{j} + 3\hat{k})$

$= (-8 - 3)\hat{i} + (5 + 6)\hat{j} + (9 - 9)\hat{k} = -11\hat{i} + 11\hat{j}$

$\therefore \|\vec{C}\| = \sqrt{(-11)^2 + (11)^2} = 11\sqrt{2}$ length unit.

Example 6

If $\vec{A} = -\hat{i} + 2\hat{j} + m\hat{k}$ and $\vec{B} = \frac{1}{3}\vec{A}$ is a unit vector , find : m

Solution

$\because \vec{B} = \frac{1}{3}\vec{A}$ $\quad\therefore \vec{B} = -\frac{1}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{m}{3}\hat{k}$

, $\because \vec{B}$ is a unit vector $\quad\therefore \|\vec{B}\| = 1$

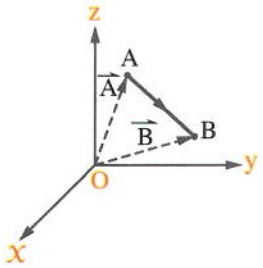
$\therefore \left(-\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{m}{3}\right)^2 = 1$ $\quad\therefore \frac{1}{9} + \frac{4}{9} + \frac{m^2}{9} = 1$

$\therefore \frac{m^2}{9} = 1 - \frac{5}{9} = \frac{4}{9}$ $\quad\therefore m^2 = 4$

$\therefore m = \pm 2$

- **Expressing a directed line segment in the space in terms of the coordinates of its terminals :**

Let A (A_x , A_y , A_z) and B (B_x , B_y , B_z) be two points in the space , their position vectors with respect to the origin are $\overrightarrow{OA}$ and $\overrightarrow{OB}$ respectively , then the directed line segment from A to B is $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$



i.e. $\boxed{\overrightarrow{AB} = \vec{B} - \vec{A}}$

For example :

If $\vec{A} = (-3 , 4 , 2)$, $\vec{B} = (5 , 0 , 7)$, then

$\overrightarrow{AB} = \vec{B} - \vec{A} = (5 , 0 , 7) - (-3 , 4 , 2) = (8 , -4 , 5)$,

$\overrightarrow{BA} = \vec{A} - \vec{B} = (-3 , 4 , 2) - (5 , 0 , 7) = (-8 , 4 , -5)$

i.e. $\boxed{\overrightarrow{AB} = -\overrightarrow{BA}}$

- **The unit vector in the direction of a given vector :**

If $\vec{A} = (A_x , A_y , A_z) \in \mathbb{R}^3$, then the unit vector in the direction of the vector $\vec{A}$ is denoted by $\overrightarrow{U_A}$ and given by the relation : $\boxed{\overrightarrow{U_A} = \frac{\vec{A}}{\|\vec{A}\|}}$

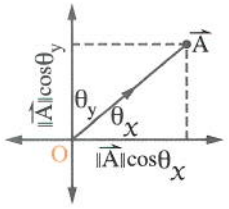
For example :

If $\vec{A} = (-2 , 2 , -1)$, then $\|\vec{A}\| = \sqrt{(-2)^2 + (2)^2 + (-1)^2} = 3$

$\therefore$ The unit vector in the direction of $\vec{A} = \overrightarrow{U_A} = \frac{\vec{A}}{\|\vec{A}\|} = \frac{(-2 , 2 , -1)}{3} = \left(\frac{-2}{3} , \frac{2}{3} , \frac{-1}{3}\right)$

Remember that

You have studied the polar form of a vector in the coordinate plane $\mathbb{R}^2$ which is $\vec{A} = (\|\vec{A}\| , \theta)$ where θ is the angle between $\vec{A}$ and the positive direction of x-axis and so $\vec{A} = \|\vec{A}\| \cos\theta \hat{i} + \|\vec{A}\| \sin\theta \hat{j}$ and if θ_x is the angle between $\vec{A}$ and x-axis and θ_y is the angle between $\vec{A}$ and y-axis , then $\sin\theta_x = \cos\theta_y$ and so the cartesian form of the vector $\vec{A}$ $\vec{A} = \|\vec{A}\| \cos\theta_x \hat{i} + \|\vec{A}\| \cos\theta_y \hat{j}$



• Direction angles and direction cosines of a vector in the space :

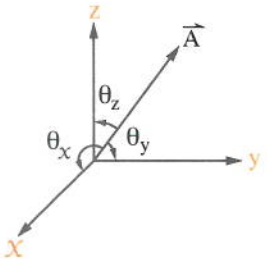
* The direction angles of the vector $\vec{A}$ in the space are the measures of angles $(\theta_x , \theta_y , \theta_z)$ that the vector makes with the positive directions of x , y , z axes respectively. Where $\theta_x , \theta_y , \theta_z \in [0 , \pi]$
* The direction cosines of the vector $\vec{A}$ in the space are the cosines of the direction angles of the vector $\vec{A}$

i.e. $(\cos\theta_x , \cos\theta_y , \cos\theta_z)$

$\therefore \vec{A} = \|\vec{A}\| \cos\theta_x \hat{i} + \|\vec{A}\| \cos\theta_y \hat{j} + \|\vec{A}\| \cos\theta_z \hat{k}$

$= \|\vec{A}\| (\cos\theta_x \hat{i} + \cos\theta_y \hat{j} + \cos\theta_z \hat{k})$

$\therefore \dfrac{\vec{A}}{\|\vec{A}\|} = \cos\theta_x \hat{i} + \cos\theta_y \hat{j} + \cos\theta_z \hat{k}$



, $\because \dfrac{\vec{A}}{\|\vec{A}\|}$ is a unit vector in direction of $\vec{A}$

$\therefore \cos\theta_x \hat{i} + \cos\theta_y \hat{j} + \cos\theta_z \hat{k}$ is a unit vector in direction of $\vec{A}$

$\therefore \cos^2\theta_x + \cos^2\theta_y + \cos^2\theta_z = 1$

Remarks

* The direction angles of a vector $\overrightarrow{AB}$ (not passing through the origin) in the space are the measures of the direction angles made by another vector passing through the origin parallel to $\overrightarrow{AB}$
* The direction cosines of any vector are the components of its unit vector in its direction.

For example : The direction cosines of the vector $\vec{A} = \left(\dfrac{A_x}{\|\vec{A}\|} , \dfrac{A_y}{\|\vec{A}\|} , \dfrac{A_z}{\|\vec{A}\|}\right)$

* The direction cosines of the positive directions of x , y , z axes or any vector in their directions are $(1 , 0 , 0) , (0 , 1 , 0) , (0 , 0 , 1)$ respectively.

* The direction angles of the + ve x , y , z axes or any vector in their directions are $(0° , 90° , 90°) , (90° , 0° , 90°) , (90° , 90° , 0°)$ respectively.

* If $(\theta_x , \theta_y , \theta_z)$ are the direction angles of the vector $\vec{A}$, then $(\pi - \theta_x , \pi - \theta_y , \pi - \theta_z)$ are the direction angles of the vector $-\vec{A}$

• Generalization :

If $(\theta_x , \theta_y , \theta_z)$ are the direction angles of the vector $\vec{A}$, then :

1. $(\theta_x , \theta_y , \theta_z)$ are the direction angles of the vector $k\vec{A}$ where $k > 0$
2. $(\pi - \theta_x , \pi - \theta_y , \pi - \theta_z)$ are the direction angles of the vector $k\vec{A}$ where $k < 0$

* If the vector $\vec{A}$ makes equal angles with the positive coordinate axes.

i.e. $\theta_x = \theta_y = \theta_z = \theta$, then $\cos\theta_x = \cos\theta_y = \cos\theta_z = \cos\theta$

, $\because \cos^2\theta + \cos^2\theta + \cos^2\theta = 1$ $\quad\therefore 3\cos^2\theta = 1$

$\therefore \cos^2\theta = \frac{1}{3}$ $\quad\therefore \cos\theta = \frac{1}{\sqrt{3}}$

$\therefore \theta \simeq 54° 44\grave{} 8\grave{}\grave{}$ $\quad$ **Or** $\cos\theta = -\frac{1}{\sqrt{3}}$

$\therefore \theta \simeq 125° 15\grave{} 52\grave{}\grave{}$

* The sum of measures of two direction angles is greater than or equal to 90°

* If the measure sum of two directions angles is 90° then the measure of the third angle is 90°

Example 7

Find the direction cosines and the direction angles of the vector : $\vec{A} = (2 , -2 , 2\sqrt{2})$

Solution

$\because \|\vec{A}\| = \sqrt{(2)^2 + (-2)^2 + (2\sqrt{2})^2} = 4$

$\because \overrightarrow{U_A} = \frac{\vec{A}}{\|\vec{A}\|} = \frac{(2 , -2 , 2\sqrt{2})}{4}$

$\therefore \cos\theta_x = \frac{2}{4} = \frac{1}{2}$, $\cos\theta_y = \frac{-2}{4} = -\frac{1}{2}$, $\cos\theta_z = \frac{2\sqrt{2}}{4} = \frac{\sqrt{2}}{2}$

$\therefore$ The direction cosines of the vector $\vec{A}$ are $(\frac{1}{2} , -\frac{1}{2} , \frac{\sqrt{2}}{2})$

, the direction angles of the vector $\vec{A}$ are $(60° , 120° , 45°)$

Example 8

If $(135°, 60°, \theta)$ are the direction angles of the vector $\vec{A}$, find : θ and if $\|\vec{A}\| = 6$, find $\vec{A}$

Solution

$\because$ $(135°, 60°, \theta)$ are the direction angles of the vector $\vec{A}$

$\therefore \cos^2 135° + \cos^2 60° + \cos^2 \theta = 1$ $\quad\quad$ $\therefore \frac{1}{2} + \frac{1}{4} + \cos^2 \theta = 1$

$\therefore \cos^2 \theta = \frac{1}{4}$ $\quad\quad$ $\therefore \cos \theta = \frac{1}{2}$ and so $\theta = 60°$

Or $\cos \theta = -\frac{1}{2}$ and so $\theta = 120°$

$\therefore$ The direction angles of the vector $\vec{A}$ are $(135°, 60°, 60°)$ or $(135°, 60°, 120°)$

$\therefore \vec{A} = 6 \cos 135° \hat{i} + 6 \cos 60° \hat{j} + 6 \cos 60° \hat{k} = -3\sqrt{2}\hat{i} + 3\hat{j} + 3\hat{k}$

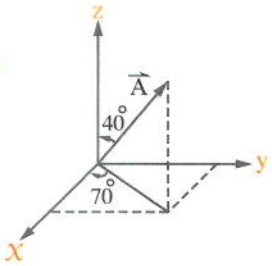
Or $\vec{A} = 6 \cos 135° \hat{i} + 6 \cos 60° \hat{j} + 6 \cos 120° \hat{k} = -3\sqrt{2}\hat{i} + 3\hat{j} - 3\hat{k}$

Example 9

The opposite figure represents a vector $\vec{A}$ whose norm is 10 units.

1. Express the vector $\vec{A}$ in algebraic form (Cartesian components)

2. Find the measures of the direction angles of $\vec{A}$



Solution

1. Resolve $\vec{A}$ into two components : the first is in the direction of $\overrightarrow{OZ}$ with magnitude

$$A_z = \|\vec{A}\| \cos \theta_z = 10 \cos 40° \simeq 7.66$$

The second is in xy-plane : $A_{xy} = \|\vec{A}\| \sin \theta_z = 10 \sin 40° \simeq 6.428$

Now, resolve the component A_{xy} into two components : the first is in the direction of $\overrightarrow{OX}$ with magnitude.

$A_x = A_{xy} \cos 70° = 6.428 \cos 70° \simeq 2.2$

The second is in the direction of $\overrightarrow{OY}$ with magnitude

$A_y = A_{xy} \sin 70° = 6.428 \sin 70° \simeq 6.04$

So, the cartesian form of the vector $\vec{A}$ is :

$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k} = 2.2\hat{i} + 6.04\hat{j} + 7.66\hat{k}$

2. To find the measures of the direction angles , we get the unit vector in the direction of $\vec{A}$

$$\overrightarrow{U_A} = \frac{\vec{A}}{\|\vec{A}\|} = \frac{1}{10}\left(2.2\,\hat{i} + 6.04\,\hat{j} + 7.66\,\hat{k}\right) = 0.22\,\hat{i} + 0.604\,\hat{j} + 0.766\,\hat{k}$$

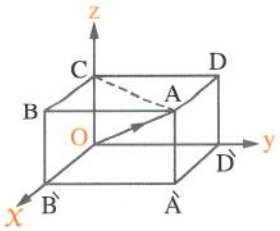
$\therefore \cos\theta_x = 0.22$, then $\theta_x = \cos^{-1}(0.22) \simeq 77.3°$

$\therefore \cos\theta_y = 0.604$, then $\theta_y = \cos^{-1}(0.604) \simeq 52.84°$

$\therefore \cos\theta_z = 0.766$, then $\theta_z = \cos^{-1}(0.766) \simeq 40°$

Example 10

The opposite figure represents a cuboid whose dimensions (DC = 8 length unit , BC = 6 length unit , OC = 5 length unit) Force $\vec{F}$ of magnitude 200 newton acts in the direction of the diagonal $\overrightarrow{OA}$



1. Express the force $\vec{F}$ in algebraic form.

2. Find the measures of the direction angles of the force $\vec{F}$

Solution

$\because \vec{A} = 6\,\hat{i} + 8\,\hat{j} + 5\,\hat{k}$

$\therefore \|\vec{A}\| = \sqrt{36 + 64 + 25} = 5\sqrt{5}$ length unit.

$\therefore$ The unit vector in direction of $\vec{A} = \frac{\vec{A}}{\|\vec{A}\|} = \frac{6}{5\sqrt{5}}\,\hat{i} + \frac{8}{5\sqrt{5}}\,\hat{j} + \frac{1}{\sqrt{5}}\,\hat{k}$

$\therefore \vec{F}$ in algebraic form is expressed as follows :

$\vec{F}$ = 200 (the unit vector in direction of $\vec{A}$)

$$= 200\left(\frac{6}{5\sqrt{5}}\,\hat{i} + \frac{8}{5\sqrt{5}}\,\hat{j} + \frac{1}{\sqrt{5}}\,\hat{k}\right)$$

$$= 48\sqrt{5}\,\hat{i} + 64\sqrt{5}\,\hat{j} + 40\sqrt{5}\,\hat{k}$$

, $\because$ the direction cosines of the vector $\vec{A} = \left(\frac{6}{5\sqrt{5}}, \frac{8}{5\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$

$\therefore$ The measures of the direction angles of the force $\vec{F}$ = (57° 32` 37``` , 44° 18` 45``` , 63° 26` 6```)